Aryaman Kaprekar

B.Tech Hons.| Cyber Physical Systems | Minor: Computational Intelligence
+91-9920690051| arkap0104@gmail.com | LinkedIn | GitHub | Website

Summary

I am driven by the curiosity to understand why a system works, where it breaks, and how far it can be pushed. This plays out in both published literature and in models I've shipped onto real hardware under real constraints. The two complement each other: research sharpens the questions I ask, and deployment keeps me honest about the answers.

Experience

- **DeepGrid Semi Pvt Ltd** *May 2025 – July 2026*
  AI Intern → ADAS Engineer → AI Team Lead
  - Led the AI team, setting research direction and technical decisions across autonomous driving programs, and represented DeepGrid Semi at the AI Impact Summit and to investors.
  - Re-architected the flagship end-to-end autonomous driving model (AD2) for edge hardware using hardware-aware optimization, operator redesign, and FP8 quantization-aware training, cutting compute cost 95% while preserving accuracy.
  - Built the full embedded deployment pipeline (PyTorch → ONNX → TensorRT → CUDA → C++) with multi-sensor fusion, raising Jetson Orin Nano throughput 240× (0.05 → 12 FPS) and cutting model size 86% (1.1 GB → 150 MB).
  - Validated on Cityscapes, NuScenes, NuPlan, IDD, and Waymo (open-loop) and CARLA/NavSim (closed-loop), achieving a Drive Score of 86 on Bench2Drive, and co-designed a 28 nm autonomous driving SoC with embedded and RTL teams.
- **Advanced Healthcare Lab, MAHE** *Dec 2024 - Current*
  Research Fellow
  - Focus areas: lightweight architectures, uncertainty quantification, quantitative explainable AI, and ablation studies for Computer-Aided Diagnostics.
  - **(Author, Scopus Q1)** *A multi-level attention CNN-transformer based framework for the detection of brain tumor using regional dual-score explainability. Nature Scientific Reports (Springer), 2026*
  - **(Co-Author, Scopus Q1)** *TrustNet: A Lightweight Network with Integrated Uncertainty Quantification and Quantitative Explainable AI for Ischemic Stroke Detection in CT Images. Nature Scientific Reports (Springer), 2026*
  - **(Co-Author, Scopus Q2)** *Automated Identification of Left Ventricular Hypertrophy using Cardiac Ultrasound Imaging: A Systematic Review of AI-Driven Approaches. Elsevier — Informatics in Medicine Unlocked, 2025*

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech Hons.	Manipal Institute of Technology, Manipal	8.84	2022-2026
Senior Secondary	Maharashtra State Board	82.67%	2022
Secondary	ICSE Board	96.16%	2020

Selected Personal Projects

- **Road-scenes Segmentation (2025):** Literature review on segmentation; lightweight hybrid GAN model achieving 63% mIoU / 72% mPA on Cityscapes with only 11.3M params.
- **Consumer Product Cost Prediction (2025):** Multimodal price prediction fusing ResNet50 + DistilBERT embeddings into an XGBoost regressor; SMAPE of 47.5.
- **End-to-end MLOps Pipeline (2025):** Production-grade pipeline with MLFlow, DVC, Docker, and Git-based CI/CD — fully reproducible from raw data to deployed model.
- **RL-based Batch Reactor Controller (2025):** RL agent for temperature regulation using Policy Mirror Descent and TD learning for coolant/heater control.
- **Generative Models for Reflection Removal (2024):** Autoencoder/GAN pipelines for single-image reflection removal; 89.77% mAP on a custom dataset.

Skills

- **Programming Languages**: Python, C++, C, Embedded C
- **Frameworks & Tools**: PyTorch, ONNX, TensorRT, CUDA, OpenCV, MLFlow, DVC, Docker, Git, Linux, MATLAB, CMake, NumPy, Pandas
- **Technical Domains**: Deep Learning, Computer Vision, Model Optimisation, Reinforcement Learning, Edge-AI, Medical Image Analysis, Digital Image Processing, IoT, Computer Architecture, Computer Networks

Certifications and Extra-Curricular

- **NPTEL & Coursera:** Top 5% in India across multiple AI/Computer Vision NPTEL courses; Deep Learning Specialization (deeplearning.ai).
- **Qualcomm Vision X:** Finalist for the reflection-removal project at IIT-B.
- **Workshops:** IoT/ML training at Bolt-IoT; Gesture Robotics at IIT-B.
- **Other:** NCC regimental training and social service; Everest Base Camp trek (5,364m); Class Representative.